

rules do not allow for any spending increases in retirement, even when markets do well, which allows for more remaining wealth when initial withdrawal rates are the same. The modified RMD rule, which increases the withdrawal rate over time to match declining longevity, is the most efficient at spending down assets over retirement. It is the only rule with a decline in remaining real wealth at the ninetieth percentile. At the median, Zolt's glide path rule maintains the largest remaining wealth on account of spending decreases without any possibility for increases. At the tenth percentile, remaining wealth is the largest for Vanguard's floor-and-ceiling rule.

In terms of changes in spending at the ninetieth percentile, the modified RMD rule provided the largest spending increase. This was followed by the ratcheting rule and the Guyton and Klinger decision rules. Meanwhile, at the median, the ratcheting rule offered the largest spending increase of 77 percent after thirty years. This was followed by the modified RMD rules.

The only strategy to experience a decline in spending at the median was Zolt's glide path spending rule. At the tenth percentile of outcomes, the modified RMD rule was the only strategy that allowed for a spending increase even in these poor market environments. Spending was 6 percent higher at the tenth percentile after thirty years.

In terms of strategies that experience the biggest reductions to spending, Vanguard's floor-and-ceiling percentages strategy experienced a 44 percent decline in spending at the tenth percentile, while Zolt's glide path strategy experienced a 41 percent decline in spending.

The next column (year 30 spending relative to baseline) shows the year thirty spending for a strategy relative to the initial spending from the baseline constant inflation-adjusted spending strategy. For this particular exhibit, all were strategies recalibrated to an initial 4 percent spending rate rather than using a PAY Rule. Thus, for this exhibit, the spending relative to the baseline strategy is the same as the spending for the strategy itself. Again, this is because all strategies in this particular exhibit start at retirement with a 4 percent withdrawal rate.